

Master Team Project Fall 2024

FindFul

Team 1

Member Name	Role
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Samrat Bikram Shah	Backend Lead and Github Master
Annika Kristin Kaul	Frontend Lead
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Milestone 1

13/11/2024

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1. Executive Summary:

In the complex landscape of student housing, international students face significant challenges when searching for affordable, convenient, and suitable accommodations. Findful emerges as an innovative digital platform designed to transform the student rental experience in Fulda, Germany, addressing critical pain points for both students and landlords.

International and local students encounter numerous obstacles when seeking suitable housing: language barriers, limited local knowledge, complex rental processes, and the overwhelming task of finding affordable, quality accommodations. Traditional housing platforms are often generic, and do not cater specifically to student needs. Findful bridges this critical gap by creating a user-friendly, comprehensive housing marketplace tailored exclusively to students' unique requirements.

Our platform offers a holistic solution that goes beyond a simple display of property listings. By providing verified student and landlord profiles, comprehensive map-based property listings, and an integrated messaging system, Findful simplifies the entire housing search process. One of our standout features is the ability for students to apply for shared accommodation as a group, rather than as individuals. This is particularly beneficial in Fulda, where shared living arrangements are common among students. This feature allows students in Fulda to easily find and apply for shared housing, and also enables them to showcase their profiles to landlords, making it easier to find the perfect match. Overall, FindFul will facilitate students in exploring detailed property information, utilizing advanced search and filtering capabilities, and creating detailed profiles that help landlords understand their background, study details, and personal characteristics, making it easier for landlords to find suitable tenants.

The market potential for Findful is significant, driven by increasing international student enrollment and the growing demand for transparent, accessible housing solutions. Our platform represents more than just a technological solution – it is a scalable model that can potentially transform student housing searches in university cities across Germany and beyond.

We are a passionate group of software development students at Fulda University, united by a shared vision to solve a critical challenge faced by students. Comprised of five talented individuals from different academic and cultural backgrounds, each team member brings unique strengths to the project: some have experienced the challenges of international student housing, while others contribute deep technical skills in software development, user experience design, and system architecture. With a mix of technical knowledge, entrepreneurial thinking, and a deep understanding of student needs, our team is committed to transforming the student housing search experience.

2. Personae and Main Use cases:

Persona 1: Ram (International Student)

Background:

Ram is an incoming master's student from Nepal who has recently enrolled in an English-language business master's program in Fulda, Germany. He comes from a very modest economic background. He doesn't know German very well, nor does he know anyone in Germany. Ram is introverted and reserved, with basic technological skills.

Goals:

- He requires secure, affordable, and furnished accommodation near the university before classes begin in April.
- He wants to navigate the housing market in a foreign country with minimal local connections and language barriers.

Skills:

- He has ideas on basic internet navigation and the usage of standard web applications. However, he has limited experience with online housing platforms.

Pain Points:

- Having tight budget constraints makes it challenging for him to find affordable housing.
- Understanding and navigating German housing websites is difficult due to language barriers and a lack of local connections in Germany.
- He is unable to transport his personal furniture, requiring him to find a furnished accommodation.

Persona 2: Tim (Guest)

Background:

Tim is a German resident from Hessen, living in his own home near Frankfurt with his wife and two children. His daughter, Lily, is about to finish her high school and is considering attending university in Fulda. Lily wants to study in a smaller, quieter city compared to Frankfurt and finds Fulda to be a good option for a peaceful academic environment. He knows it's not practical to travel back and forth from his home to the university almost everyday for the classes.

Goals:

- Find a high-quality, spacious single-room apartment for his daughter near the university in Fulda.
- Ensure the accommodation is safe, comfortable, and close to campus to avoid long commutes.
- He doesn't care about the price of the room nor the furnishings.

Skills:

- Proficient in using technology and familiar with online real estate platforms.

Pain Points:

- Fulda is not confirmed as Lily's final choice yet, but he wants to prepare by scouting suitable accommodations.
- Since price and furnishings are not concerns, he finds it challenging to identify apartments that offer the right amount of space and comfort.
- He is unable to transport his personal furniture, requiring him to find a furnished accommodation.

Persona 3: Thomas Hartmann (Landlord)

Background:

Thomas is a real estate owner and landlord from Gießen, Germany. Together with his wife and daughter, he lives in a small house in the suburbs of Gießen. He specializes in renovating and renting out apartments to students in Hessen. Since 2005, Thomas has bought and renovated three apartment complexes in Fulda. He has recently finished renovating a building near the Fulda University of Applied Sciences.

Goals:

- Rent out all his apartments to students in Hessen.
- Plans to expand his business by buying and renovating new buildings to rent to students.
- Earn money for his family to provide them with a good life.

Skills:

- Good at constructing and renovating apartments.
- Basic understanding of technology and online real estate platforms.

Pain Points:

- Does not have a lot of time, as he has to drive across all of Hessen daily to maintain his multiple apartment complexes.
- New technologies overwhelm him. He prefers simple structures that allow him to reach his goals efficiently.

Persona 4: Anna Kaiser (Moderator)

Background:

Anna is a young entrepreneur from Wächtersbach, near Fulda, Germany. After her studies in International Business Management, she started working as a marketing manager and moderator at FindFul. She is an energetic, outgoing person who is passionate about her work. FindFul is not only her job; it is her passion project, as she has many international friends who had problems finding accommodations during their studies. When she helped her friends find accommodations, she realized that there were too many unserious and unhelpful listings on the market, which is why she decided to address that issue herself.

Goals:

- Help international students with an easy process to find suitable accommodations..
- Reduce poor listings of accommodations on the market.

Skills:

- Anna is very proficient in utilizing software and online tools. She has strong interpersonal skills and a good understanding of human behavior.

Pain Points:

- She doesn't like to use outdated web applications.
- She is frustrated by internet scams and unverified listings that waste her time and her friends' time when searching for accommodations.

Use Cases:

- 1. Search for Housing - Tim (Guest):** Tim opens the FindFul platform as a guest to search for affordable apartments. He filters the listings, specifying that the room should be a single-room apartment, over 25 square meters in size, and in close proximity to the university. Tim finds several options that meet his criteria, including both furnished and unfurnished choices. He is pleased to see that all the rooms are spacious, bright, and close to the university, giving him confidence that there are really good options available.
- 2. View Selected Housing - Ram (Student):** After searching for apartments within his budget of 380 euros, Ram opens two accommodations closest to the university in new tabs to compare them. As he examines each listing, he reviews detailed information about each room, including its location, existing tenants, and additional photos. He notices that one room is furnished, while the other is not, which helps him make a decision on which room to choose.
- 3. Connect with Landlords - Ram (Student):** After identifying a suitable furnished room from the list of rooms, Ram uses the platform's messaging feature to contact the landlord directly. He inquires about the availability, terms, and any additional costs.
- 4. Apply as a Group - Ram (Student):** Ram is about to move to Fulda with two friends. Ideally, they want to apply for a room together, as having familiar faces when moving to a new country can be helpful. On FindFul, after finding a suitable room for three people, they create a group of three to apply together. They then apply for the room as a group rather than individually and communicate with the landlord as a group.
- 5. Add Listing of New Real Estate - Thomas (Landlord):** Thomas has recently finished renovating a new apartment complex in Fulda. He now visits the portal to add a listing for the new building. He inserts the basic information about the building, details about the different apartments available, and adds photos to the portal. After a few steps, Thomas successfully submits the new listing.
- 6. Reviewing New Listings - Anna (Moderator):** Anna begins her daily review of new listings of real estates that should be posted to FindFul. She goes through the list one by one. She proofreads and inspects the submitted texts and images of the first listing and approves it. Anna continues with the second listing. The listing shows inconsistent images, so Anna declines the posting of the listing.

3. List of main data items and entities:

1. **Student** represents a verified Fulda University student who must complete email verification (hs-fulda.de) before registration.
2. **Landlord** is any individual or organization offering rental properties on the platform.
3. **Guest** is an unauthenticated visitor with limited platform access. They can browse public listings, view basic property information, and see location details, but cannot contact landlords or access full listing features.
4. **Moderator** is a system maintainer with comprehensive moderation powers. They review all new property listings and associated media, with the authority to approve or reject content based on platform guidelines.
5. **Real estate** is a representation of physical property available for rental, specifically targeting student accommodation in Fulda.
6. **Rental Listing** is the central data element representing a property available for rent. It contains all property details including title, description, monthly rent, and property type.
7. **Location** is a geographical description of a property's position, showing its location relative to important places like the university and city center.
8. **Media Item** is visual or image content associated with a rental listing, serving as a critical representation of the property's physical appearance and characteristics.

4. Initial list of functional requirements:

User registration and authentication

1. Users can register for an account as either a student or landlord. Before registration, users have guest status on the platform.
2. Users can log into their accounts using passkeys, offering a secure and convenient alternative to traditional passwords.
3. Users who are registering as a student are required to verify their student status to complete registration.
4. The system allows registered users to log out of their accounts.

Profile Management

5. Students can manage their profile including personal information, contact details, and upload profile photos.

Managing Listings

6. Landlords can post detailed accommodation listings with information such as rent, room type, amenities, location, and availability.
7. Each listing can include multiple media files for a clear visual presentation.
8. Landlords can edit or remove their accommodation listings as necessary.

9. Landlords can view student cards for any listing to which students have applied, providing a comprehensive idea regarding the students.

Admin functions

10. Admins have the authority to approve or reject listings based on their relevance. Listings will only be visible to students after receiving admin approval.
11. Admins can ban users who violate the platform's terms of service.

Search & Filtering

12. All Users on the system can perform searches for listings using keywords.
13. The system provides filters for listings, including key criteria such as their proximity to Hochschule or Bahnhof.
14. Users can explore multiple listings through an interactive map which allows them to access listing details directly from the map.

Accommodation Listings

15. Users can click on a listing to view detailed information including a description, photos, location on a map, rent breakdown and availability.
16. The system may suggest additional listings that match the user's previous search preferences or saved listings.
17. Users may be able to compare multiple listings side-by-side by key categories, such as rent, amenities, and location.

Group Application for Shared Accommodation

18. Students can make a group by adding up to 6 other students on the system.
19. Students can apply as a group to any rental accommodation that has available rooms for students in a group.

Communication

20. Students can contact landlords directly using in-app messaging.
21. Students and Landlords can manage their active chats in the My Messages section, for viewing conversations with each other.

5. List of non-functional requirements:

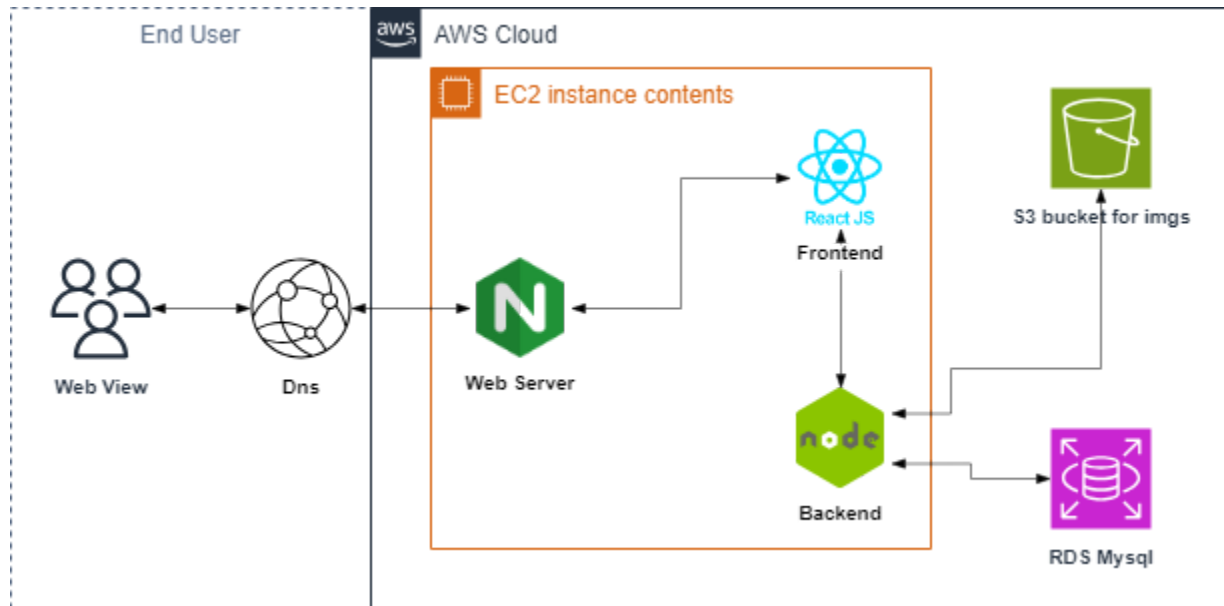
1. Application shall be developed, tested and deployed using tools and servers approved by Class CTO and as agreed in Milestone 0. Application delivery shall be from chosen cloud server
2. Application shall be optimized for standard desktop/laptop browsers e.g. must render correctly on the two latest versions of two major browsers
3. All or selected application functions must render well on mobile devices
4. Data shall be stored in the database on the team's deployment cloud server.
5. Full resolution free media shall be downloadable directly, and full resolution media for selling shall be obtained after contacting the seller/owner
6. No more than 50 concurrent users shall be accessing the application at any time
7. Privacy of users shall be protected, and all privacy policies will be appropriately communicated to the users.
8. The language used shall be English (no localization needed)
9. Application shall be very easy to use and intuitive
10. Application should follow established architecture patterns
11. Application code and its repository shall be easy to inspect and maintain
12. Google analytics shall be used (optional for Fulda teams)
13. No e-mail clients shall be allowed.
14. Pay functionality, if any (e.g. paying for goods and services) shall not be implemented nor simulated in UI.
15. Site security: basic best practices shall be applied (as covered in the class) for main data items
16. Application shall be media rich (images, video etc.). Media formats shall be standard as used in the market today
17. Modern SE processes and practices shall be used as specified in the class, including collaborative and continuous SW development
18. For code development and management, as well as documentation like formal milestones required in the class, each team shall use their own GitHub to be set-up by class instructors and started by each team during Milestone 0
19. The application UI (WWW and mobile) shall prominently display the following exact text on all pages "Fulda University of Applied Sciences Software Engineering Project, Fall 2024 For Demonstration Only" at the top of the WWW page. (Important to not confuse this with a real application).

6. Competitive analysis:

Key Feature	Wg-gesucht https://www.wg-gesucht.de/en/	Airbnb https://www.airbnb.com/	Immobilien Scout24.de https://www.immobilienscout24.de/	FindFul
Main Navigation via search	✓	✓	✓	✓
Focused on students	✓	✗	✗	✓
Embedded map	✓	✓	✓	✓
In-site messaging	✓	✓	✓	✓
Shareable student information	✗	✗	✗	✓
English support	✓	✓	✗	✓
Apply as a student group for a shared accommodation	✗	✗	✗	✓

Our application sets itself apart by allowing students to apply for rentals as groups and even find other students interested in forming groups for shared accommodation. In addition, our platform incorporates widely-used tools like map visualization combined with student-focused functionalities. These include full English support tailored for international students, along with comprehensive profiles on landlords and rental properties. Students can also create shareable profiles, giving landlords deeper insights into potential tenants and helping facilitate personality-based matches. With these features, we aim to create a highly personalized, student-friendly rental experience that makes finding suitable housing smoother, more collaborative, and better aligned with students' needs and preferences.

7. High-level system architecture and technologies used:



For the realization of this project, the proposed architecture is the following:

- **Frontend:** All the views from the app are developed in Javascript with React as the main framework.
- **Backend:** In charge of handling users and other services requests. Developed in node js with express.js as a backend framework.
- **Web Server:** Set up with Nginx, this module works as a gateway to communicate the frontend with the backend.
- **Cloud services:** We chose AWS as the project's cloud provider, mainly for the computer engine services, object storage for images and database simplification. The main services in use are:
 - **Ec2:** As the compute component, aws offers 1 vCpu and 1 Gb of ram for 750 hrs/month. It includes a dedicated ipv4 and 30 gbs of storage.
 - **S3:** S3 buckets help simplify the process of storing images and videos.
 - **RDS:** As with the S3, RDS simplify the set up and operation of our Mysql databases.

8.Team and roles:

Member Name	Role
Sarah Majeed	Team Lead and Document Master
Samrat Bikram Shah	Backend Lead and Github Master
Annika Kristin Kaul	Frontend Lead
Kemer Jimenez	Team Member
Abdul Rahman Zuhaib	Team Member

9. Checklist

Task	Status
Team found a time slot to meet (online) outside of the class	DONE
GitHub master chosen	DONE
Team decided and agreed together on using the listed SW tools and deployment server	DONE
Team ready and able to use the chosen back and front end frameworks and those who need to learn are working on learning and practicing	ON TRACK
Team lead ensured that all team members read the final M1 and agree/understand it before submission	DONE
GitHub organized as discussed in class (e.g. master branch, development branch, folder for milestone documents etc.)	ON TRACK